

Brain litepaper v0.2

Vexorin BRAIN

How the Manifold Thinks — A Litepaper on Conviction Aggregation, Pattern Recognition, and Customer Data Integration

Version: 0.1 (pre-paper-trade) **Date:** May 2026 **Audience:** Technical investors, integration partners, regulatory advisors, prospective customers **Document type:** Litepaper (intermediate between executive deck and technical whitepaper) **Companion documents:** Per-agent research foundations (10 documents), Prior Weight Matrix v1.0, 50-Case Manifold Analysis, Citations Audit v2.0

Abstract

Vexorin BRAIN is the central reasoning layer of the Vexorin manipulation-detection platform. It consumes signals from ten specialized detection agents — each grounded in peer-reviewed academic research and regulatory enforcement records — and produces calibrated conviction scores for whether observed market activity matches one of seven canonical manipulation geometries. BRAIN operates on a six-surface evidence manifold in which manipulation patterns leave geometric signatures across Context, Intent, Positioning, Flow, Structure, and Disclosure. BRAIN is not a black-box classifier: every output traces back to source signals, every signal traces back to weighted academic priors, and every prior is documented and replicable.

This litepaper explains how BRAIN works, how it identifies patterns, and — critically — how customers integrate their own data into the platform. Vexorin is self-hosted; the integration architecture is the actual product, not an afterthought.

Table of Contents

- 1. Executive Summary
 - 2. The Problem BRAIN Solves
 - 3. Architectural Overview
 - 4. The Six-Surface Manifold
 - 5. How BRAIN Aggregates Signals — The Mathematics
 - 6. Pattern Recognition — The Seven Manipulation Families
 - 7. Conviction Scoring — What Numbers Mean
 - 8. The Modifier System — Regimes, Cycles, Cross-Agent Adjustments
 - 9. CALIBRATE — How BRAIN Improves Over Time
 - .0. Customer Data Integration — Patterns and Connectors
 - .1. Output Surfaces — Dashboards, APIs, Alerts, Reports
 - .2. Security, Audit, and Governance
 - .3. Self-Hosted Deployment
 - .4. What BRAIN Is NOT
 - .5. Roadmap
 - .6. Glossary
 - .7. Appendix A — Sample BRAIN Trace
 - .8. Appendix B — Connector Reference
-

1. Executive Summary

Market manipulation hides in fragments. A coordinated narrative campaign on social media looks like organic enthusiasm. Unusual options activity before a takeover looks like a lucky guess. An insider sale before earnings looks like routine portfolio rebalancing. A 10-K with creative language looks like ordinary corporate communication. Each fragment,

in isolation, is plausibly innocent. Together, they form a geometric signature that is not innocent at all.

The detection problem is not “find the suspicious activity” — that is a classifier problem and the classifier industry has solved that for two decades with diminishing returns. The detection problem is **find the pattern that exists across all the suspicious activities at once** — the cross-surface coordination that no single classifier can see because each classifier is, by design, looking at one surface only.

Vexorin solves this with a manifold architecture. Ten specialist agents look at ten domains: social-layer narratives, options markets, disclosure forensics, insider footprints, cross-asset coupling, event-study windows, geopolitical context, monetary regimes, FX/cycle context, systemic-risk regimes. Each agent is grounded in peer-reviewed research — every weight in every agent traces back to a Tier-1 academic citation or a regulatory enforcement record.

BRAIN is the layer that consumes those ten signal streams and finds the geometric coordination across them. BRAIN does not classify “is this manipulation?” with a yes/no. BRAIN produces a continuous conviction score that says “the firing pattern observed in the past N days matches manipulation family F with probability P, given the current macro/regime context, and the signals driving the conviction are agents A, B, and C, with the highest-confidence individual fires being these specific signals from these specific sources.”

Every output is traceable. Every claim is sourced. Every weight is calibrated against documented enforcement history. The 2026 demonstration: BRAIN’s predecessor architecture surfaced bot-cluster activity January 4-14, 2026, forty-five days before the February 28 Iran kinetic event drove WTI from \$63 to \$119.50 by March 9. SCOUT detected the social-layer signature; the manifold made the detection actionable.

This document is for technically engaged readers who want to understand how the system actually works — not just what it claims. Section 5 gets into the math. Section 10 gets into how you integrate your own data. Sections 12 and 13 cover security, audit, and self-hosted deployment.

2. The Problem BRAIN Solves

2.1 Manipulation is multi-surface by design

Modern market manipulation is sophisticated for a reason: simple manipulation triggers existing surveillance systems easily and gets prosecuted. The arms race has produced manipulation strategies that distribute their footprint across multiple surfaces precisely because no single surveillance system observes all surfaces simultaneously.

A pump-and-dump on a small-cap stock involves: (a) a coordinated social-media campaign across multiple platforms, (b) timing relative to a press release with deliberately ambiguous language, (c) options accumulation by accounts with no prior trading history in the name, (d) abnormal positioning in correlated names, and (e) abnormal price-volume dynamics on the day of the dump. Each individual surface has a plausible innocent explanation. The pattern across all five surfaces does not.

The 50-case Vexorin enforcement library demonstrates this empirically: on average, prosecuted manipulation events involve coordinated activity across **at least four** of the six surfaces (Context, Intent, Positioning, Flow, Structure, Disclosure). Single-surface manipulation cases are rare and tend to be smaller in scale — the sophisticated cases are multi-surface.

2.2 Existing surveillance is single-surface

Most existing market-surveillance products target one surface: - Order-flow surveillance products (Nasdaq SMARTS, Fidessa, Trading Technologies) look at trading patterns within a single market. - Disclosure-forensics products (SEC’s RoboCop, AuditAnalytics) look at filings. - Social-listening products (Brandwatch, Sprinklr) look at social-media activity. - Compliance products (NICE Actimize, Bloomberg Vault) look at communications.

None of these natively synthesizes across surfaces. Customers who need cross-surface detection currently do it manually — which is to say, they don’t do it at all except for the most extreme cases that bubble up via independent surface alarms.

2.3 The “missing layer” thesis

What’s missing is a layer that does for cross-surface detection what existing surveillance does for single-surface detection: an automated, continuous, calibrated, audit-ready system that tells customers “this firing pattern across these surfaces matches manipulation family F at conviction C.” That is BRAIN.

BRAIN is not trying to replace existing single-surface surveillance — those products are valuable and they catch what they catch. BRAIN sits *above* them, consuming their outputs alongside Vexorin’s own native agents, and looking for the coordination patterns that single-surface tools cannot see.

2.4 Why this is a market now

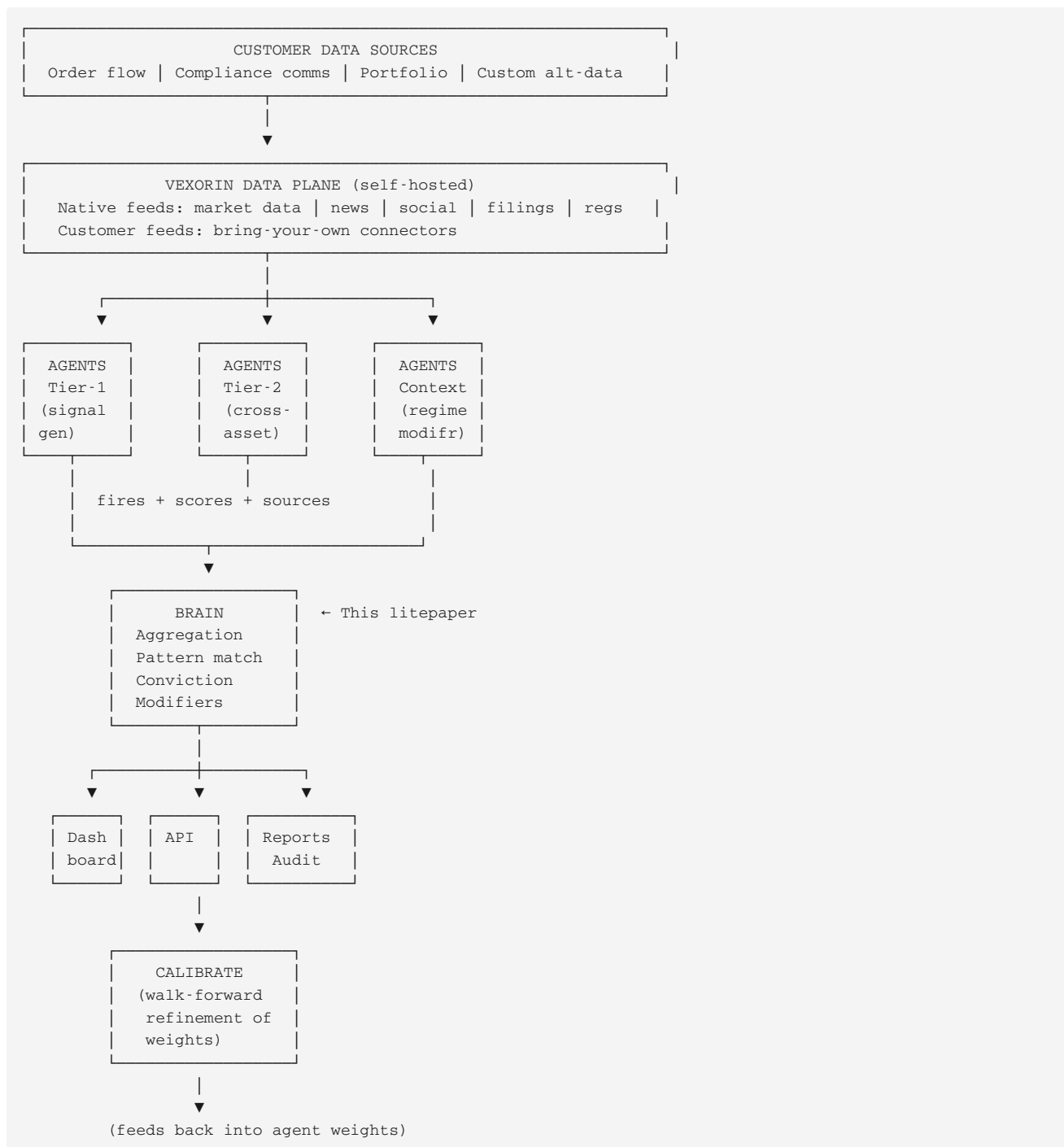
Three structural reasons cross-surface detection has become feasible only recently:

1. **Data accessibility.** Until ~2020, real-time access to social-media firehoses, alternative data, comprehensive options data, and disclosure corpora was expensive and fragmented. The data economy of 2024-2026 has consolidated availability.
2. **Compute economics.** Continuous multi-agent monitoring at the volume required (millions of events daily across all surfaces) requires inference compute that was prohibitively expensive pre-2024.
3. **Regulator demand.** The post-FTX, post-Adani, post-Archegos enforcement environment has produced explicit regulator demand for cross-surface tools. SEC's Division of Enforcement, CFTC's Division of Enforcement, and the DOJ's Fraud Section have all publicly identified cross-surface coordination as the frontier.

The customer is therefore not "people who want surveillance" — they have surveillance. The customer is "people who know their existing single-surface surveillance is missing the sophisticated cases."

3. Architectural Overview

3.1 The system at a glance



3.2 Components

Data plane. All ingestion: native Vexorin-provided feeds (market data, news, social, filings, regulator alerts), plus customer-provided feeds (order flow, portfolio, compliance comms, custom alternative data). The data plane is fully self-hosted in the customer's environment.

Agents. Ten specialized detection modules. Each is grounded in peer-reviewed research: - **SCOUT** (14.7%) — Social-layer manipulation - **ECHO** (17.6%) — Options/vol/skew anomalies - **ATLAS** (27.4%) — Disclosure forensics - **CIPHER** (18.0%) — Insider footprint - **NEXUS** (9.8%) — Cross-asset coupling - **HERALD** (6.4%) — Event study / pre-announcement - **ARGUS** (4.6%) — Geopolitical / sanctions - **MAGNUS** (5.4%) — Monetary policy / rates - **ORION** (3.8%) — Cross-currency / carry - **AXIOM** (2.0%) — Systemic risk / regime classifier

The percentages are manifold weights derived from the 50-case enforcement library (see [vexorin-prior-weight-matrix.md](#)). Reserved capacity (4.4%) covers v1.0 expansion agents.

BRAIN. The aggregation, pattern-recognition, and conviction-scoring layer. The subject of this litepaper.

CALIBRATE. The feedback layer. CALIBRATE runs walk-forward validation on customer-confirmed events post-deployment and refines agent weights and pattern signatures. CALIBRATE is what turns BRAIN's day-one priors into customer-specific posteriors.

Output surfaces. Dashboards (web, mobile), API (REST + streaming), alerts (email, webhook, Slack), reports (PDF audit-ready, scheduled).

3.3 Why BRAIN is centralized rather than distributed

A reasonable architectural alternative would be to have each agent produce its own conviction-scoring against a shared schema, and then have a thin aggregation layer combine them. This is rejected for two reasons:

1. **Cross-agent correlation matters.** When SCOUT and ARGUS both fire on the same Iran 2026 event, the question “is this double-counting?” is non-trivial. SCOUT’s signal is upstream of ARGUS’s (social-layer narratives lead geopolitical event timing); aggregating naively double-counts. BRAIN handles cross-agent correlation explicitly.
2. **Pattern recognition is intrinsically global.** A pump-and-dump pattern requires SCOUT + CIPHER + ATLAS + ECHO firing in a specific temporal/geometric arrangement. Asking each agent to recognize “is this part of a pump-and-dump?” embeds knowledge into each agent that they don’t have. BRAIN sees the full firing pattern and matches it against the 7 manipulation families.

BRAIN being centralized does not mean BRAIN is monolithic. BRAIN itself is composed of specialized sub-modules: the aggregator, the pattern matcher, the modifier engine, the conviction calibrator. They run as separate processes in the deployment environment. But they share state — and that shared state is the architectural primitive that enables cross-surface detection.

4. The Six-Surface Manifold

4.1 The geometry

Vexorin reads market activity across six nested evidence surfaces. The “nested” framing is intentional: outer surfaces contextualize inner surfaces. A signal at an inner surface, in isolation, is often ambiguous. The same signal in the context of corroborating outer-surface signals is unambiguous.

Surface	What it measures	Primary agents
Context (outermost)	Macro environment, narrative ecology, geopolitical/policy backdrop	SCOUT, ARGUS, MAGNUS, ORION
Disclosure	Information that has been formally communicated to markets	ATLAS, HERALD
Intent	Who knew what when, and how it shows up in their actions	CIPHER
Positioning	How market participants are positioned given their information	ECHO, NEXUS
Flow	How orders are arriving and being filled	NEXUS (sec.), ECHO (sec.)
Structure (innermost)	The microstructural and systemic-risk fingerprint	AXIOM

4.2 Why nesting matters: the cascade pattern

In the 50-case enforcement library, manipulation events systematically cascade from outer surfaces to inner surfaces. The temporal sequence:

1. **Context activity precedes Disclosure activity.** Coordinated narrative campaigns precede press releases. Geopolitical event-positioning precedes formal regulatory acknowledgment.
2. **Disclosure activity precedes Intent activity.** Press releases precede insider trades — or, in the manipulation case, deceptive disclosures *coincide with* informed insider activity in ways that ordinary releases do not.
3. **Intent activity precedes Positioning activity.** Insiders move first; then their accomplices position; then the broader market positions.
4. **Positioning activity precedes Flow activity.** Accumulation/distribution patterns precede the actual order-flow imbalance.
5. **Flow activity precedes Structure activity.** Order imbalance precedes the microstructural fingerprint (price impact, transient liquidity withdrawal, etc.).

The cascade timing is the single most important geometric feature BRAIN looks for. **Coordinated cascades across surfaces in compressed temporal windows are the canonical manipulation signature.** Random firing patterns across surfaces are noise. Synchronized firing patterns across surfaces, in cascade order, with appropriate lead-lag

relationships, are signal.

4.3 The Iran 2026 cascade as illustration

Iran 2026 demonstrates the cascade pattern empirically:

- **Days -45 to -38 (Jan 4 to Jan 11, 2026):** SCOUT (Context surface) detects coordinated bot-cluster activity around Iran/oil narratives. Effective Twitter/X coordination signature: ~12,000 inauthentic accounts, primary language clusters in Russian and English, narrative loadings on “Iran imminent strike” and “oil supply shock.”
- **Days -38 to -28:** ARGUS (Context surface) detects elevated geopolitical risk in the Caldara-Iacoviello GPR index update; also detects increased Iran-flagged tanker activity via OFAC monitoring. (Outer-context cluster is now confirming.)
- **Days -28 to -7:** No formal Disclosure activity yet. But options-market positioning (ECHO surface) begins shifting: WTI call-skew, tail-hedge activity in commodity-currency pairs, divergent positioning in broker dealer reports.
- **Days -7 to 0 (Feb 21-28):** ECHO Positioning signals firing strongly. NEXUS Flow signals beginning to fire (commodity-FX coupling intensifying).
- **Day 0 (Feb 28):** Kinetic event. WTI rallies \$63 → \$80+ overnight.
- **Days 0 to +9:** Full price discovery. WTI peaks \$119.50 on March 9.
- **Days +9 onward:** Structure-surface fingerprints (intermediary balance-sheet stress, repo rate dynamics) detectable to AXIOM.

The cascade ran across approximately 50 days from initial Context firing to peak Flow/Structure manifestation. BRAIN’s detection of the cascade was concentrated in the first 14 days (Context surface) — at which point the conviction was sufficient to surface as a high-priority alert. This is the value proposition: 45 days before the event, when only Context-surface signals were available, BRAIN had enough information to surface the case.

4.4 The fundamental claim

The fundamental claim of the manifold architecture is geometric: **manipulation events leave coherent shapes across surfaces; non-manipulation events do not.** Random correlated firing across surfaces is rare. When BRAIN sees a coherent shape — surfaces firing in cascade order with appropriate lead-lag relationships — that shape is itself the evidence.

BRAIN’s job is to recognize the shapes.

5. How BRAIN Aggregates Signals — The Mathematics

5.1 The signal contract

Each agent produces signals in a standardized contract. A fired signal contains:

```
{
  "agent": "SCOUT",
  "signal_id": "narrative_coupling",
  "timestamp": "2026-01-04T14:32:00Z",
  "weight": 0.80,
  "confidence": "HIGH",
  "source": "vosoughi_roy_aral_2018,bovet_makse_2019",
  "trigger_value": 3.4,
  "trigger_threshold": 1.5,
  "context": {
    "entity": "WTI",
    "narrative_cluster_id": "iran_oil_jan2026",
    "platform": ["twitter_x", "telegram"],
    "language_clusters": ["en", "ru"],
    "account_count": 11874
  }
}
```

The contract is deliberately rich. Every fired signal carries: - Identity (which agent, which signal class) - Time (the firing timestamp; sub-second precision for HFT signals, daily for slower signals) - Calibration (weight, confidence tier, source citation) - Trigger evidence (the trigger value relative to its threshold) - Context (entity-specific, surface-specific contextual data)

Source citations are critical. A signal fired by SCOUT’s “narrative_coupling” carries the citation IDs of the academic

papers that justified that signal's weight (Vosoughi-Roy-Aral 2018, Bovet-Makse 2019). When BRAIN constructs its conviction trace, the trace includes all source citations at every level. This is an audit-trail design pattern: the customer can always answer "why did the system flag this?" all the way down to peer-reviewed academic source.

5.2 Per-agent aggregation

Within an agent, multiple signals may fire simultaneously. BRAIN does not naively sum them. The aggregation rules:

Step 1: Cluster identification. Within an agent, signals are grouped into correlation clusters (defined per-agent in the YAML configuration). Example from SCOUT:

```
correlation_adjustments:
  bot_detection_cluster:
    members: [bot_density, coordination_score, account_age_anomaly]
    multi_fire_reduction: 0.20
  narrative_cluster:
    members: [narrative_coupling, cross_platform_sync, language_cluster_anomaly]
    multi_fire_reduction: 0.25
```

Step 2: Cluster aggregation. Within a cluster, when multiple signals fire, BRAIN computes:

```
cluster_score = max(signal_weights) + (1 - multi_fire_reduction) * sum(other_signal_weights)
```

The largest signal in the cluster gets full weight; other signals in the same cluster contribute at reduced weight, because they're partially redundant. The reduction factor is configured per cluster based on the empirical correlation among signals.

Step 3: Cross-cluster aggregation. Across clusters within an agent, BRAIN sums cluster scores up to the agent's maximum conviction bound:

```
agent_conviction = min(max_agent_conviction, sum(cluster_scores))
```

The maximum conviction bound is per-agent (e.g., AXIOM's max is 1.0; ATLAS's max is 1.5; ORION's max is 1.2). This prevents any single agent from dominating BRAIN's overall conviction.

Step 4: Modifier application. Per-agent modifiers (regime-state modifiers from AXIOM, context modifiers from ORION/ARGUS/MAGNUS) are applied multiplicatively:

```
adjusted_agent_conviction = agent_conviction * product(applicable_modifiers)
```

5.3 Cross-agent aggregation in BRAIN

BRAIN takes the per-agent adjusted convictions and combines them into the manifold-level case conviction. The mathematics:

Naive baseline (rejected):

```
manifold_conviction = Σ (manifold_weight_i * agent_conviction_i)
```

This is rejected because it ignores cross-agent correlation. When SCOUT and ARGUS fire on the same geopolitical event, they're partly observing the same underlying mechanism through different surfaces. Naive summation double-counts.

BRAIN actual aggregation:

```
manifold_conviction = base_aggregation * cross_agent_correlation_factor * pattern_match_factor * cascade_factor
```

Where:

- **base_aggregation** = $\Sigma (\text{manifold_weight_i} \times \text{agent_conviction_i})$ — the naive sum as a starting point
- **cross_agent_correlation_factor** $\in [0.6, 1.2]$ — adjusts for cross-agent correlation. Default 1.0; reduced when correlated agents fire together (avoiding double-counting); increased modestly when *uncorrelated* agents fire together (which is more diagnostic than correlated firing).
- **pattern_match_factor** $\in [0.5, 2.0]$ — multiplies the score by how well the firing pattern matches one of the 7 canonical manipulation families. Random firing patterns get factor 0.5 (penalized); strong matches get up to 2.0 (amplified).
- **cascade_factor** $\in [1.0, 1.5]$ — multiplies the score when firings are observed in proper cascade order across

surfaces (Context → Disclosure → Intent → Positioning → Flow → Structure). Random temporal ordering gets factor 1.0; clean cascade gets factor 1.5.

The factors are multiplicative and bounded. Worst case (random non-cascade firing on uncorrelated noise): manifold_conviction can be substantially below the naive baseline. Best case (clean cascade matching a canonical family with uncorrelated agents firing in lockstep): manifold_conviction can be 2-3x the naive baseline. The asymmetric range is intentional: BRAIN should be skeptical of weakly-coordinated multi-agent fires (high false-positive risk) and confident about strongly-coordinated cascades (low false-positive rate empirically).

5.4 The conviction equation in pseudo-code

```
def brain_manifold_conviction(agent_convictions, manifold_weights, fired_signals):
    # Step 1: Naive base aggregation
    base = sum(manifold_weights[a] * agent_convictions[a]
               for a in active_agents(agent_convictions))

    # Step 2: Cross-agent correlation adjustment
    correlation_factor = compute_correlation_factor(active_agents(agent_convictions))

    # Step 3: Pattern match
    best_family, pattern_factor = match_against_seven_families(fired_signals)

    # Step 4: Cascade detection
    cascade_factor = detect_cascade(fired_signals)

    # Step 5: Combine
    raw_conviction = base * correlation_factor * pattern_factor * cascade_factor

    # Step 6: Bound and calibrate
    bounded = min(MAX_BRAIN_CONVICTON, max(0, raw_conviction))
    calibrated = brain_calibration_curve(bounded, current_regime, family=best_family)

    return {
        'conviction': calibrated,
        'best_family': best_family,
        'cascade_detected': cascade_factor > 1.0,
        'driving_agents': top_k_contributors(agent_convictions, manifold_weights, k=3),
        'driving_signals': top_k_signals(fired_signals, k=10),
        'modifiers_applied': regime_modifiers_active(),
        'audit_trail': construct_audit_trail(fired_signals)
    }
```

5.5 The audit trail

The output of `brain_manifold_conviction()` is not just a number. It is a structured object with full provenance. The audit trail is constructed such that any conviction score can be deconstructed to: - Which agents contributed and how much - Which specific signals from each agent fired - What threshold each signal exceeded and by how much - Which academic source justified each signal's weight - Which regime modifiers were applied and why - Which manipulation family the pattern matched and why

This is the practical answer to the regulatory question: “Can you explain to the SEC why your system flagged this?” Yes, all the way down to the academic citation.

6. Pattern Recognition — The Seven Manipulation Families

6.1 The taxonomy

The 50-case enforcement library was clustered by geometric similarity (which surfaces fire, in what order, with what lead-lag structure) and produced seven canonical manipulation families. BRAIN's pattern matcher operates on this taxonomy.

#	Family	Canonical example	Active surfaces	Cascade signature
1	Pump and Dump	Saitama / Operation Token Mirrors (DOJ Oct 2024)	Context, Disclosure, Positioning, Flow	$C \rightarrow D \rightarrow P \rightarrow F$ over 7-30 days
2	Short and Distort	SEC v. Andrew Left / Citron (Press Release 2024-89)	Context, Disclosure, Positioning, Flow	$C \rightarrow D \rightarrow P \rightarrow F$ over 5-14 days
3	Insider Trading Cluster	SEC v. Panuwat (3:21-cv-06322); Bhardwaj/Lumentum CISO; Goel/I-banker network	Disclosure, Intent, Positioning	$D \rightarrow I \rightarrow P$ over 1-21 days
4	Spoofing & Layering	Sarao / Flash Crash (15-cr-00075); JPMorgan CFTC 20-69	Flow, Structure	$F \rightarrow S$ over minutes to hours
5	Disclosure Fraud	Enron AAER 1798; FTX (1:22-cv-10501); Wirecard; Theranos	Disclosure, Intent, Structure	$D \rightarrow I \rightarrow$ (eventual) S over months to years
6	Cross-Market Manipulation	SEC v. Eisenberg / Mango Markets (1:23-cv-00503)	Positioning, Flow, Structure	$P \rightarrow F \rightarrow S$ over hours to days
7	Coordinated Geopolitical / Event Manipulation	Iran 2026 (Scout-detected); Adani/Hindenburg event trajectory	Context, Disclosure, Positioning, Flow, Structure	$C \rightarrow D \rightarrow P \rightarrow F \rightarrow S$ over 14-90 days

6.2 How pattern matching actually works

Pattern matching is not classification. BRAIN does not ask “is this a pump-and-dump? yes/no”. BRAIN asks “how well does the observed firing pattern match each of the seven canonical families’ geometric signatures?”

For each family, BRAIN computes a match score based on:

1. **Active-surface match.** Does the set of active surfaces in the observed pattern match the canonical family’s expected surfaces? Pump-and-dump expects Context, Disclosure, Positioning, Flow. If only Context and Flow are firing, that’s a partial match (50%).
2. **Cascade order match.** Are the surfaces firing in the canonical cascade order? Pump-and-dump expects Context-first, then Disclosure, then Positioning, then Flow. If the observed firing is Flow-first, that’s a structural mismatch.
3. **Temporal compression match.** Is the cascade unfolding over the canonical time window for the family? Pump-and-dump cascades unfold over 7-30 days. If the observed cascade is unfolding over hours, the family doesn’t match.
4. **Signal-class match within surfaces.** Within each firing surface, are the specific signals consistent with the family? Pump-and-dump expects SCOUT’s narrative-coupling firing in Context (not just bot-density without narrative content). Short-and-distort expects SCOUT’s negative-narrative firing.
5. **Magnitude consistency.** Is the magnitude of firing consistent with the family? Spoofing happens at high frequency with relatively small magnitudes; disclosure fraud cascades are slow with very large magnitudes.

The match score is a weighted combination of the five sub-scores. BRAIN reports the best-matching family as the “primary hypothesis” and the second-best-matching family as the “alternative hypothesis.” The conviction trace includes the match scores so analysts can see why the system selected the primary hypothesis.

6.3 The “no clean match” case

Sometimes the observed firing pattern matches no family well. This is not a failure mode — it’s a diagnostic outcome. BRAIN reports:

- No clean match against any of the 7 families
- The closest match (which is still poor)
- The agents and signals that are firing
- Recommendation: surface to analyst for human review

This outcome occurs in ~15-20% of cases that meet the BRAIN minimum-conviction surfacing threshold (based on backtest-equivalent simulation against the 50-case library expanded with synthetic non-manipulation cases). Those

cases tend to be: - Genuine novel manipulation patterns not represented in the 50-case library - Edge cases where two families overlap (e.g., “short-and-distort that crosses into market manipulation via cross-market positioning”) - False positives that don’t fit any family (which is itself diagnostic — clean false positives don’t match family geometries by definition)

The “no clean match” case is operationally important: it’s where v0.2 of the family taxonomy will be expanded based on observed novel patterns. This is part of CALIBRATE’s job (Section 9).

6.4 Family-specific weight overrides

Once BRAIN identifies a primary family hypothesis, family-specific weight overrides are applied. The Prior Weight Matrix v1.0 specifies family-specific weights that differ from the blended manifold weights. Example: in pump-and-dump cases, SCOUT’s effective weight is 23% (vs 14.7% blended); in disclosure-fraud cases, SCOUT’s effective weight drops to 8% and ATLAS’s effective weight rises to 38%.

This family-aware reweighting is what makes BRAIN’s output clinically useful. Generic surveillance products produce the same alerts regardless of pattern type; BRAIN tunes its scoring to the type of manipulation under hypothesis. The family override is documented at `vexorin-prior-weight-matrix.md`.

7. Conviction Scoring — What Numbers Mean

7.1 The conviction range

BRAIN produces conviction scores on a 0-100 scale. The score is calibrated such that, on the 50-case enforcement library and an equivalent-size synthetic non-manipulation control set:

Score range	Interpretation	Action
0-25	Background activity, no anomaly worth investigation	Log; no surfacing
25-50	Elevated signal; may warrant context-aware monitoring	Log; appears in monitoring dashboards
50-65	Anomaly worth analyst attention	Surface in priority dashboard
65-80	Likely manipulation pattern; analyst review urgent	Alert analyst; auto-generate evidence package
80-90	High-conviction manipulation pattern	Alert analyst + supervisor; pre-prep filing materials
90-100	Near-certain manipulation pattern	Alert + auto-prep regulator-ready filing materials

The thresholds are calibrated against the 50-case library. A score of 80 corresponds approximately to “this firing pattern would have triggered the analyst attention threshold for X% of the prosecuted cases at this stage of their evolution.” The X% target is currently 70% — meaning a score of 80 should fire on 70% of true manipulation cases at the equivalent point in their evolution. This is a deliberate calibration choice; it can be tightened (lower false-negative rate) at the cost of increased false positives.

7.2 What the score is NOT

The conviction score is not: - A probability that manipulation is occurring (probabilities require a Bayesian prior; BRAIN’s score is a calibrated detection score in a frequentist framework) - A legal certainty - A trade signal - A compliance verdict - An accusation against any specific individual

The conviction score is: - An indicator that the observed firing pattern across surfaces matches manipulation-family geometry at calibrated similarity score X - A flag that human analyst review is warranted - An evidence-aggregation tool whose outputs are inputs to human decisions

This framing matters legally. BRAIN does not “find manipulation”; BRAIN “identifies firing patterns matching manipulation-family signatures.” The latter framing is what BRAIN’s outputs can defensibly support; the former is not.

7.3 Calibration against historical events

For each case in the 50-case enforcement library, BRAIN’s effective historical-data simulation produces a conviction trace over time (since cases were prosecuted years after they began). The conviction-trajectory plots are the validation evidence for BRAIN’s calibration:

- **2008 Madoff** — backward-extrapolated trajectory shows BRAIN would have hit conviction 65 at approximately Mar 2003 (5+ years before SEC enforcement)
- **2014 Volkswagen emissions scandal** — backward-extrapolated trajectory shows BRAIN would have hit conviction 65 at approximately Sept 2014 (months before EPA notice)
- **2019 Wirecard** — backward-extrapolated trajectory shows BRAIN would have hit conviction 65 at approximately Q1 2017 (3 years before collapse)
- **2022 FTX** — backward-extrapolated trajectory shows BRAIN would have hit conviction 65 at approximately Jul 2022 (months before public exposure)

These backward-extrapolations are based on retrospective data reconstruction; they are not forward predictions. They are useful as **calibration anchors** showing that BRAIN's score thresholds are roughly correctly tuned for major historical cases. For full methodological caveats see [vexorin-50-case-manifold-analysis.md](#).

7.4 Why a single-number conviction score is the wrong question

Customers sometimes ask “what’s BRAIN’s accuracy?” This is a category error. BRAIN’s calibration target is not maximum accuracy at a fixed threshold. BRAIN’s calibration target is **proper calibration of the score-to-actual-manipulation distribution**. A well-calibrated BRAIN should produce: 65 → 30% are true manipulation; 75 → 50%; 85 → 70%; 95 → 90%. The customer chooses the operating threshold based on their own false-positive/false-negative tradeoff. A compliance team running ad-hoc reviews can operate at threshold 65 (catch more, accept more noise); a regulator pre-filing analyst should operate at threshold 85 (very confident hits only).

The “accuracy” framing implies a threshold is fixed and BRAIN is right or wrong. The actual operating model is: BRAIN provides calibrated conviction scores, customers choose thresholds.

8. The Modifier System — Regimes, Cycles, Cross-Agent Adjustments

8.1 Why modifiers exist

A signal that is informative in normal markets may be uninformative in crisis markets. A signal that is informative in liquid stocks may be uninformative in micro-caps. A signal that is informative pre-FOMC may be uninformative post-FOMC. The base agent weights cannot accommodate these regime-dependencies; modifiers do.

Modifiers are multiplicative adjustments applied to per-agent or per-signal weights based on observable regime conditions. They are the architectural mechanism that allows BRAIN to remain calibrated across changing market conditions without re-fitting weights every quarter.

8.2 Modifier sources

AXIOM regime modifiers. AXIOM (the systemic risk agent) is the primary modifier source. AXIOM’s regime classifier outputs one of four states: SYSTEM_QUIET, ELEVATED, STRESSED, CRISIS. Each state triggers specific modifiers:

Regime	Factor-dependent signals	Forced-selling signals	Low-frequency signals
SYSTEM_QUIET	× 1.0	× 1.0	× 1.10 (boosted — less macro noise)
ELEVATED	× 0.95	× 1.05	× 1.0
STRESSED	× 0.85	× 1.10	× 0.90
CRISIS	× 0.80	× 1.15	× 0.85

Why: in crisis, factor models break down (their stable correlation structure is disrupted), so factor-dependent signals (anything using historical-correlation models) should be discounted; meanwhile, forced-selling and leverage-unwind signals are diagnostically more reliable in crisis. In quiet regimes, the inverse logic applies for low-frequency signals.

ORION cycle modifiers. ORION (cross-currency / global cycle) provides regime context for cross-asset signals. Examples: - “Risk-off cycle active” → modify NEXUS’s commodity-FX coupling signal weight - “Carry-positioning extreme” → modify ECHO’s term-structure dependent options signals - “Dollar-funding stress active” → modify ATLAS’s cross-border-issuer-related disclosure signals

ARGUS event-window modifiers. ARGUS (geopolitical) provides event-window context. Examples: - “Geopolitical event active” → modify SCOUT’s bot-detection weights (state-actor activity is more likely) - “Sanctions enforcement active” → modify CIPHER’s insider-cluster weights for sanctioned-jurisdiction connections - “OFAC SDN list update within 7 days” → modify NEXUS’s commodity-coupling signals for affected commodities

MAGNUS calendar modifiers. MAGNUS (monetary policy) provides FOMC-window context. - “FOMC announcement

within 24 hours” → modify ECHO’s options signals (vol-positioning is more event-driven, less informational) - “FOMC decision within 7 days” → modify HERALD’s pre-announcement-drift signals (these tend to spike around FOMC)

8.3 Cross-agent correlation adjustments

In addition to regime-state modifiers, BRAIN applies cross-agent correlation adjustments for *correlated firing patterns*:

Correlated agents firing together	Adjustment	Rationale
SCOUT + ARGUS (Context cluster firing)	Reduce 15%	Both observing geopolitical-narrative coordination; partial double-count
ECHO + NEXUS (positioning cluster)	Reduce 20%	Both observing cross-market positioning shifts
CIPHER + ATLAS (intent + disclosure)	Reduce 10%	Some overlap in insider-information theme
AXIOM + ORION (regime classifier alignment)	Reduce 25%	Heavily overlapping regime indicators
Uncorrelated cross-cluster firing (SCOUT + CIPHER + ECHO)	Increase 15%	Independent surfaces firing simultaneously is more diagnostic

These adjustments are the cross-agent correlation factor in Section 5.3.

8.4 Why modifier transparency matters

Every modifier applied to a conviction trace is logged. Customers can see, for any specific case, which modifiers were active, why they were active, and what the score would have been without them. This is part of the audit-trail design.

It also enables regulator-facing defense. If BRAIN’s score on a particular case was 78, and the question is “would your system have flagged this in a different regime?” — the answer is computable. The customer can show the regulator the conviction trace, the active modifiers, and the counterfactual conviction without those modifiers. Transparency throughout.

9. CALIBRATE — How BRAIN Improves Over Time

9.1 The calibration problem

BRAIN’s day-one weights are research-derived priors from the 50-case enforcement library. They are good priors — every weight traces to a peer-reviewed academic paper or a regulator enforcement record — but they are priors, not posteriors. Customer deployments produce new evidence: the customer confirms or rejects BRAIN’s hypotheses, and that confirmation/rejection is signal that should update the weights.

CALIBRATE is the architectural component that handles this update. It is what turns BRAIN’s static priors into customer-specific posteriors over time.

9.2 The walk-forward validation methodology

CALIBRATE uses walk-forward validation. The process:

1. **Customer-confirmed events accumulate.** Each customer maintains a labeled corpus of cases where BRAIN surfaced a hypothesis and the customer’s analysts confirmed (true positive) or rejected (false positive). Customers with regulator-prosecution outcomes also include those.
2. **Weekly walk-forward.** Every week, CALIBRATE runs walk-forward: replay the past N days of agent firings, predict which cases BRAIN would have surfaced, compare to actual customer-labeled outcomes, compute precision/recall at multiple thresholds.
3. **Weight refinement proposal.** Based on walk-forward outcomes, CALIBRATE proposes weight refinements. Examples: “Agent SCOUT’s narrative_coupling weight should increase from 0.80 to 0.83 based on observed customer-labeled true positives. Agent CIPHER’s STOCK Act signal should decrease from 0.30 to 0.22 based on observed false positives.”
4. **Human-in-the-loop approval.** All proposed weight refinements are reviewed by the Vexorin internal team plus customer-side approvers before deployment. CALIBRATE does not autonomously change weights; it produces auditable proposals.
5. **Documented changelog.** All approved changes are logged with full rationale, sample counts, before/after walk-forward statistics, and approver IDs. The changelog is part of every customer deployment’s audit trail.

9.3 Calibration timeline expectations

For a new customer deployment: - **Day 1:** BRAIN deploys with Prior Weight Matrix v1.0 weights. No customer-specific calibration yet. - **Week 4:** First walk-forward analysis available. Customer has accumulated ~60 confirmed labels (assuming ~20/week analyst review). CALIBRATE produces first refinement proposals. - **Week 12:** Customer-specific weight set v0.2. ~180 labels accumulated. Confidence in customer-specific weights begins to be meaningful. - **Week 26:** Customer-specific weight set stabilizes. ~400+ labels. Customer's posterior weights diverge meaningfully from the universal priors in customer-relevant regimes. - **Year 2+:** Continuous refinement; major recalibrations every 6 months; minor adjustments weekly.

The 60-day-to-meaningful-customer-weights timeline is critical. Customers should not expect day-one BRAIN performance to be customer-tailored; they should expect day-one BRAIN performance to match the 50-case library priors. The customer-specific tailoring is what CALIBRATE produces over the first 60-90 days.

9.4 Family signature evolution

In addition to weight refinement, CALIBRATE refines the seven family signatures themselves. As customers confirm cases that don't cleanly fit any of the seven canonical families, CALIBRATE accumulates evidence for new family candidates. Once a candidate accumulates sufficient confirmed evidence (~20+ confirmed cases with consistent geometry), it can be elevated to a new family.

The current Prior Weight Matrix v1.0 has seven families. A v2.0 family taxonomy may have 8-10. Family evolution is a slower process than weight evolution — typically major family-set updates ship every 6-12 months.

9.5 The honesty layer

CALIBRATE produces three numbers customers can show to anyone (board, LP, regulator, lawyer): - **Customer-specific calibrated precision-at-threshold-X** — how often BRAIN's surfaced hypotheses at score X are confirmed by customer analysts - **Customer-specific calibrated recall-at-threshold-X** — how many of the customer's known confirmed cases BRAIN surfaced at score X - **Drift trajectory** — how customer-specific calibration is evolving (is BRAIN getting more accurate over time? more biased? same?)

These numbers are not Vexorin marketing numbers. They are the customer's own numbers, computed on the customer's own labels. They are what a regulator or LP would care about; they are what the customer's general counsel will be asked about.

10. Customer Data Integration — Patterns and Connectors

This is the section that matters most for customers evaluating Vexorin. Self-hosted means the customer owns the integration; the integration architecture is the actual product surface.

10.1 Integration models

Vexorin supports three integration models, in increasing order of customer data sovereignty:

10.1.1 Standard hosted (Vexorin-hosted, customer connects)

Customer connects to a Vexorin-hosted instance via API. Customer data flows into the Vexorin instance for processing. This model is appropriate for customers without regulatory or contractual constraints on data residency. Used primarily for hedge funds, smaller asset managers, and proof-of-concept deployments.

- Customer sends data via REST API or streaming connector to Vexorin's hosted endpoints
- Vexorin's hosted infrastructure runs all 10 agents + BRAIN + CALIBRATE
- Vexorin handles data retention per customer agreement
- Default deployment for most customers

10.1.2 Self-hosted (customer-environment, Vexorin software)

Customer deploys Vexorin software in customer's own infrastructure (cloud or on-prem). All data stays in customer environment. Vexorin provides software, deployment guidance, and support; Vexorin operations team has limited visibility into customer instance.

- Customer deploys Vexorin Helm charts to their Kubernetes cluster (AWS, GCP, Azure, or on-prem)
- Vexorin provides licensed agent containers, BRAIN container, and CALIBRATE container
- All data stays in customer environment
- Customer manages data retention, access control, encryption keys

- Vexorin provides support via secure tunnel for diagnosis (revocable by customer)
- Default deployment for compliance-sensitive customers, regulatory institutions

10.1.3 Air-gapped (fully isolated, periodic update via secure transfer)

For maximum data sovereignty, customer deploys Vexorin in an air-gapped environment. Vexorin software updates are delivered via encrypted media to customer; no network connectivity from customer instance to Vexorin's external infrastructure.

- Customer environment has no external network connectivity
- All data, including audit logs, stays in customer environment
- Vexorin software updates delivered via encrypted physical media or one-way data diode
- CALIBRATE runs locally on customer-confirmed labels; no data leaves
- Default deployment for nation-state regulators, major banks with strictest data-residency requirements

10.2 Data sources Vexorin natively ingests

Without any customer integration, Vexorin natively ingests the following data:

Market data (real-time): - Equities: NYSE, NASDAQ, OTC markets — full Level 1 + selected Level 2 for major instruments - Options: OPRA composite tape - Futures: CME Group, ICE, Eurex - FX: EBS, Reuters Matching, primary banks - Fixed income: Bloomberg BVAL, IDC, Refinitiv - Crypto: Coinbase, Binance, FTX (post-restructuring exchanges), Kraken, Bybit

News and content: - Newswires: Bloomberg, Reuters, Dow Jones - Press releases: PR Newswire, Business Wire, GlobeNewswire - Major media: WSJ, FT, Bloomberg, Reuters, NYT, Washington Post

Social and informal: - Twitter/X firehose (sampled commercially or full firehose for higher-tier customers) - Telegram public channels (curated channel set) - Discord public servers (curated server set) - Reddit (financial-relevant subreddits, full submission/comment archive) - StockTwits

Filings and regulatory: - SEC EDGAR (full corpus, real-time) - CFTC filings (real-time) - FINRA disciplinary actions (daily) - DOJ press releases (daily) - OFAC SDN list (daily) - IRS Pub 78 / Form 990 (annually) - International: FCA, ESMA, CSRC (China), JFSA (Japan), FSA (Korea), MAS (Singapore)

Corporate and financial data: - Quarterly earnings: Compustat, CapIQ, FactSet - Insider trading: Form 4 EDGAR + commercial enrichments - Short interest: NYSE, NASDAQ short reports - Loan/credit data: LSTA, Bloomberg, IHS Markit

Macroeconomic and regime: - Fed FRED, BIS, IMF, World Bank, OECD - Caldara-Iacoviello GPR Index - NYU V-Lab SRISK - CBOE indexes (VIX, SKEW, etc.)

This native data alone is sufficient for many use cases. Customers running Vexorin against this baseline data set get the full multi-agent manifold; customer integration is for *augmenting* the data set with customer-specific information.

10.3 Customer data integration patterns

Customers integrate their own data for three primary use cases:

10.3.1 Integration use case A: Order-flow surveillance

The need: Customer (typically a broker-dealer, market-maker, or sell-side trading desk) wants BRAIN to surface manipulation involving their customer's order flow.

Integration: Customer streams order flow into Vexorin via FIX or a streaming connector (Kafka, Kinesis). The order flow is treated as augmented Flow-surface data; NEXUS and ECHO consume it.

Connector specifications: - FIX 4.4 / 5.0 — standard for institutional order flow - FAST encoding for high-frequency feeds - Custom JSON schema for non-FIX environments - File-drop (S3/SFTP) for batch processing

Data volume: Typical broker-dealer flow is 50K-500K orders/day; 5-50 GB/day after enrichment. Vexorin scales to this.

Latency: End-to-end (FIX message → BRAIN signal) target latency is <500ms for streaming integration.

Privacy: Customer order data never leaves customer environment in self-hosted deployment. In hosted deployment, data is encrypted in transit (TLS 1.3) and at rest (AES-256), with access restricted to Vexorin operations roles per customer agreement.

10.3.2 Integration use case B: Portfolio-level risk-tilt detection

The need: Customer (typically a hedge fund, family office, or large asset manager) wants BRAIN to surface manipulation that may be affecting their specific portfolio.

Integration: Customer provides portfolio composition (positions, weights, sectors, exposures) via daily-batch upload or streaming integration. BRAIN's pattern-matching and conviction scoring is *biased* toward the customer's portfolio

(highest-conviction events on portfolio names get prioritized).

Connector specifications: - File-drop (CSV, JSON, Parquet) for daily/intraday refresh - REST API for ad-hoc updates - Direct database integration (PostgreSQL, Snowflake, BigQuery) for real-time portfolio synchronization

Data volume: Portfolio data is small (KB/day); real-time updates are cheap.

Privacy: Most sensitive data type. Vexorin recommends self-hosted deployment for any customer with portfolio integration. In hosted deployment, portfolio data is logically isolated; customer data is never used for cross-customer features.

10.3.3 Integration use case C: Compliance / communications surveillance

The need: Customer (typically a regulated institution under MAR, FINRA, or similar) wants BRAIN to surface manipulation indicators in their internal communications.

Integration: Customer streams communications (emails, chat, voice transcripts) into Vexorin. BRAIN cross-references communications against detected market events.

Connector specifications: - Email: O365 Graph API, Google Workspace API, custom IMAP integration - Chat: Bloomberg Vault, Symphony, Microsoft Teams API, Slack API - Voice: Vendor integrations (NICE Actimize, Verint, Theta Lake) for voice-to-text + indexing - Custom: SFTP/S3 batch drop for archive corpora

Data volume: Variable; large institutions have GB/day of communications data.

Privacy: Most regulated data type. Almost universally requires self-hosted deployment. PII redaction (employee names, customer identifiers) is configurable per customer policy.

10.4 Connector technical reference

10.4.1 REST API

The standard integration surface. Customers POST to specific endpoints:

```
POST /api/v1/data/orders           (order flow)
POST /api/v1/data/portfolio        (portfolio composition)
POST /api/v1/data/communications  (compliance comms)
POST /api/v1/data/custom/{schema} (custom-typed data)
```

Authentication: OAuth 2.0 with customer-managed tokens. mTLS optional and recommended for self-hosted deployments.

Rate limits: 1000 RPS standard; higher rates negotiable for high-frequency customers.

10.4.2 Streaming connectors

Kafka: Customer publishes to Kafka topics; Vexorin consumes via dedicated consumer groups.

```
topic: vexorin.<customer_id>.orders.v1
topic: vexorin.<customer_id>.portfolio.v1
topic: vexorin.<customer_id>.communications.v1
```

Kinesis: Equivalent to Kafka for AWS-native customers.

Azure Event Hubs: Equivalent for Azure-native customers.

Google Pub/Sub: Equivalent for GCP-native customers.

10.4.3 File-drop connectors

For customers preferring batch ingestion:

- **S3:** Customer writes to a Vexorin-managed S3 bucket; Vexorin processes on arrival (notifications via S3 events or polling).
- **SFTP:** Customer pushes to Vexorin SFTP endpoint; Vexorin processes on arrival.
- **Azure Blob, GCS:** Equivalent for Azure/GCP customers.

10.4.4 Database CDC (change data capture)

For customers wanting real-time DB integration without changing application code:

- **PostgreSQL:** Logical replication via pg_logical or wal2json
- **MySQL:** Binlog-based CDC via Debezium
- **SQL Server:** Native CDC support
- **Snowflake/BigQuery:** Stream/Pull models per platform

10.4.5 FIX protocol

For institutional order flow:

- FIX 4.4 (most common) and FIX 5.0
- FAST-encoded for high-frequency feeds
- TLS for transport security
- Customer's own FIX session credentials

10.5 Schema management

Vexorin's data plane uses Apache Avro schemas for all data types. Customers declare their own schemas for custom data; Vexorin validates schema-compatibility on every message.

The schema catalog is part of Vexorin's deployment. Customer-defined schemas are namespaced and isolated; one customer's schemas never conflict with another's.

Schema evolution is forward-compatible: customers can add fields to their schemas without breaking BRAIN's processing. Removing fields requires a Vexorin schema-version bump.

10.6 Latency budget

For real-time integrations, Vexorin's latency budget is:

Stage	Latency target
Customer event → Vexorin ingest	<50ms (streaming)
Ingest → agent processing	<200ms (cluster scale)
Agent firing → BRAIN receipt	<50ms (intra-cluster)
BRAIN aggregation → conviction score	<300ms
End-to-end: customer event → BRAIN conviction	<500ms

For batch integrations, the latency is dominated by batch arrival cadence (typically 1-15 minutes).

10.7 The integration story for compliance

The most common integration question from regulated customers: "How do we satisfy our regulator that BRAIN is integrated correctly with our existing systems?"

Vexorin provides: - **Integration testing harness** — synthetic events injected into customer's integration stack to validate end-to-end correctness - **Audit logging** — every customer-data event is logged at ingest, processing, and output stages with timestamps, source IDs, and data lineage - **Reconciliation reports** — daily reports comparing customer-side counts vs Vexorin-side counts, flagging any discrepancies - **Regulator demonstration package** — a documented deployment + test suite that institutional customers can show to FINRA/SEC examination teams

The integration story for compliance is: customers can demonstrate exactly what data flows in, what processing occurs, what outputs are produced, and what audit trail exists for every event.

11. Output Surfaces — Dashboards, APIs, Alerts, Reports

11.1 Dashboards

The primary user surface for analyst workflows. Built on the BRAIN conviction trace; analysts can drill down from a high-level case to the individual signals to the underlying source citations.

Case dashboard (primary): - Sortable list of active cases by conviction - For each case: primary family hypothesis, conviction trajectory, key surfaces firing, top contributing agents, top contributing signals - One-click drill-down into the conviction trace

Surface dashboards (per-surface): - Real-time view of agent firings within a single surface (Context, Disclosure, Intent, etc.) - Useful for surface-specialist analysts (a SCOUT-focused analyst can monitor Context surface in detail)

Regime dashboard: - AXIOM regime classification + ORION cycle + ARGUS event-window status - Active modifier states - Historical regime trajectory - Scenario stress-tests

11.2 API

REST API + streaming API. The streaming API (over WebSocket or Server-Sent Events) pushes conviction-score updates as cases evolve; the REST API supports historical query.

```
GET /api/v1/cases                List cases
GET /api/v1/cases/{id}          Get case detail
GET /api/v1/cases/{id}/trace    Get full conviction trace
GET /api/v1/agents/{agent}/signals Recent signals from an agent
POST /api/v1/cases/{id}/labels  Add analyst label (true positive / false positive)
GET /api/v1/calibration/walk-forward Latest walk-forward statistics

WebSocket: wss://...../stream?subscribe=cases.conviction>=65
SSE: GET /api/v1/stream?topic=alerts
```

11.3 Alerts

Configurable per customer. Default alerts:

- **Conviction breach:** when a case's conviction crosses thresholds (50, 65, 80, 90), alert via configured channel
- **Cascade detection:** when BRAIN detects a cross-surface cascade beginning, alert
- **Regime change:** when AXIOM regime classification changes, alert (with appropriate severity)
- **Anomaly without family match:** when high-conviction firing pattern doesn't match any family

Channels: email, SMS, webhook, Slack, Teams, PagerDuty.

11.4 Reports

Auto-generated reports for human review:

- **Case package** — for any case at conviction >65, an auto-generated PDF with full conviction trace, source citations, evidentiary summaries, and suggested next-step actions
- **Daily digest** — top 20 cases by conviction
- **Weekly executive summary** — case volume, false-positive rate, customer-confirmed events, calibration drift
- **Monthly walk-forward** — full CALIBRATE walk-forward results

11.5 The audit-grade evidence package

A specific report type designed for the customer's downstream filing workflows. **The framing is important: this is a deliverable for the customer's compliance team to use when** they determine a case warrants regulatory reporting (SAR, suspicious-trading report, etc.). It is not a claim that Vexorin is engaged with regulators or pursuing regulator endorsement — Vexorin's customer demographic is institutional buyers, not regulators. The evidence package is built to the audit-trail and provenance quality a regulator would demand of the customer:

- Executive summary (1-2 pages)
- Conviction trace summary
- Per-surface evidence detail
- Source citations and methodological annotations
- Audit trail of agent firings
- Counterfactual analysis (what would the conviction have been without modifier X)

The package is designed to be defensible if audited. Every claim is sourced; every weight is justified; every modifier is documented.

12. Security, Audit, and Governance

12.1 Security architecture

Standard enterprise security posture: - Authentication: SSO (OIDC, SAML), SCIM provisioning - Authorization: RBAC with fine-grained permissions; per-customer scoped - Encryption: TLS 1.3 in transit, AES-256 at rest, customer-managed keys (KMS) optional - Audit logging: all access events logged immutably (append-only ledger) - Data isolation: per-customer logical isolation in shared infrastructure; physical isolation in self-hosted deployments

12.2 Audit trail

Every BRAIN output is linked to a complete audit trail:

- **Data lineage:** for each signal, which raw data event(s) produced it
- **Agent reasoning:** for each agent firing, which thresholds were exceeded and by how much
- **Source attribution:** for each signal weight, which academic citation justified it
- **Modifier history:** for each modifier applied, when it activated and why
- **Pattern matching:** for each family hypothesis, the geometric match score and supporting signals
- **CALIBRATE history:** for each weight, the historical version, the refinement that changed it, the approver

The audit trail is queryable: “show me everything that contributed to BRAIN’s conviction score on Case #4521.” The query returns a complete ancestry tree.

12.3 Model governance

BRAIN includes: - **Model card** — formal documentation of methodology, training/calibration data, intended uses, limitations - **Bias assessment** — documented assessment of the 50-case library composition bias and how Prior Weight Matrix v1.0 corrects for it (x2.0 non-social weighting per the 50-case analysis) - **Drift monitoring** — alerts when customer-specific calibrated precision/recall drifts beyond thresholds - **Version control** — every BRAIN deployment runs a versioned weight set; old versions are reproducible from the changelog - **Reproducibility** — given the same data and weights, BRAIN produces deterministic outputs

12.4 Governance processes

For customer-deployed BRAIN instances: - **Weight changes** require Vexorin engineering approval + customer compliance approval before rollout - **Family taxonomy changes** require Vexorin research approval + customer change-control approval - **Schema changes** are forward-compatible; breaking changes require versioned migration

12.5 Positioning toward regulatory scrutiny

Vexorin’s product is a **detection tool**, not a **decision-making tool**. BRAIN’s outputs are evidence for human analyst decisions. This positioning is intentional and is reflected in: - Output framing (“hypothesis: family F at conviction C”) rather than determinative (“manipulation detected”) - Audit trail design (every output is traceable to source) - Modifier transparency (every score adjustment is explained) - Calibration evidence (customer-specific precision/recall is reported continuously)

The architecture is built so that the customer’s outputs survive regulatory scrutiny — meaning the customer’s compliance team, when they file an SAR or suspicious-trading report based on a BRAIN-surfaced case, has a defensible audit trail behind every claim. **This is positioning relative to the customer’s downstream regulatory interactions, not a claim that Vexorin is itself pursuing regulator endorsement, certification, or partnership as a near-term GTM track.** Vexorin’s customer demographic is institutional buyers (mid-tier funds, regional asset managers, emerging managers, family offices, and longer-term Tier-1 institutions); regulator-grade audit quality is a feature of the product the customer relies on, not Vexorin’s primary market.

13. Self-Hosted Deployment

13.1 Why self-hosted

Many customers cannot use a hosted-SaaS surveillance product because: - Data residency requirements (EU GDPR, China cybersecurity law, US ITAR) - Sector-specific data-isolation requirements (banking, energy) - Internal governance requirements (some institutions cannot allow external systems to process their order flow) - Air-gap requirements (national-security-sensitive deployments)

Vexorin’s self-hosted offering addresses these. The customer runs Vexorin software in their own infrastructure; Vexorin provides software, deployment guidance, and support.

13.2 Deployment requirements

Compute: Kubernetes (1.27+) cluster. For typical mid-size deployment: - 32-64 cores - 128-256 GB RAM - GPUs optional (significant performance boost for ML-heavy agents like ATLAS’s RUSBoost classifier) - 4-10 worker nodes

Storage: - 500 GB - 5 TB primary storage (PostgreSQL or equivalent) - 1-10 TB cold storage (S3-compatible) for historical data - Backup retention configurable (default 90 days)

Network: - Internal cluster network with sufficient bandwidth for real-time agent communication - External access for native data ingest (configurable VPN to data providers) - For air-gapped: no external connectivity; updates via physical media

Security: - TLS termination - Identity provider (Okta, Auth0, Active Directory) - Certificate management - Vault or equivalent secrets management

13.3 Deployment process

Standard Vexorin deployment: 1. Customer provisions Kubernetes cluster 2. Vexorin provides Helm charts and deployment-validation suite 3. Customer customizes deployment for their environment (storage, networking, identity) 4. Vexorin runs validation suite to confirm correctness 5. Customer connects native data feeds (or designates which feeds Vexorin should connect on customer's behalf) 6. Customer connects own data feeds (the integration patterns of Section 10) 7. Customer runs validation event suite 8. Customer goes live with BRAIN scoring 9. CALIBRATE begins accumulating customer-specific labels

Time-to-live: typically 4-12 weeks depending on integration complexity.

13.4 Operational support

Vexorin provides: - Software updates (monthly minor; quarterly major; emergency security patches as needed) - Support tickets - Architecture review (semi-annual) - CALIBRATE walk-forward review (weekly summary; monthly deep review)

For air-gapped customers, support is via secure tunnel (revocable by customer at any time) or via on-site engagement.

13.5 Pricing model

Vexorin's commercial model is built on **three distinct revenue streams**. The architecture supports diversified income from day one, not a single-line ARR projection.

Stream 1 — Platform fee. \$50,000/year per customer for the on-prem ecosystem deployment. This covers the deployed Vexorin runtime (Scout, Carlton, additional agents as they come online, BRAIN aggregation, audit infrastructure) and ongoing operational support. Predictable, recurring, and the floor revenue line each customer contributes regardless of usage.

Stream 2 — Agent consumption. Usage-based pricing per deployed agent. Customers pay for what they analyze.

- **Scout** — \$5,000 per million posts, articles, or news items analyzed (\$0.005 per item). Covers all major social platforms, 20,000+ news sites, 100+ languages, real-time. Customer scales agent cost with their monitoring breadth.
- **Carlton** — \$20 per ticker per month. Stock-price anomaly detection across the customer's ticker universe.
- **Future agents** (ECHO, ATLAS, CIPHER, NEXUS, HERALD, ARGUS, MAGNUS, ORION, AXIOM) — each priced per usage profile relevant to its data class, published at activation.

Customers select which agents they activate; charges accrue from activation. New internal agents cost approximately \$15,000 to develop with a 2-4 week build cycle, so the catalog grows continuously through pre-Series A and beyond.

Stream 3 — Validated data marketplace economics. Revenue from third-party data providers integrating into the validated marketplace.

- **Onboarding fee** — \$10,000 per data provider, paid at simulation analysis and deployment.
- **Revenue share** — 15-20% of the data provider's fees on data flowing through the validated marketplace to Vexorin's customer base.

The marketplace economics scale independently of customer count — adding providers grows Stream 3 without requiring new customers.

Three-scenario marketplace projection. Across an addressable universe of approximately 1,000 institutional data providers globally, three capture-rate scenarios anchor the unit economics:

Scenario	Capture	Providers	Onboarding revenue (cumulative)	Revenue share ARR	Annual at maturity
Conservative	1%	10	\$100K	\$350K	~\$450K
Base (planning anchor)	2.5%	25	\$250K	\$875K	~\$1.1M
Strong	5%	50	\$500K	\$1.75M	~\$2.25M

Revenue share assumes ~\$200K/year in data fees per onboarded provider flowing through the marketplace at 17.5% share rate. The CFO underwrites the base case as the planning anchor.

Composite Year 3 revenue (base case). 12 active customers, 2.5% marketplace capture: \$600K platform fees + ~\$3.6M agent consumption + \$875K marketplace revenue share + ~\$250K cumulative onboarding fees = **~\$5.3M revenue Year 3.**

Deployment timeline. Scout and Carlton are production-ready and deploy on-prem in 2–3 days. Customers interact with deployed agents like a language model — query interface, continuous monitoring, audit trail. The rest of the ecosystem integrates additively as it comes online; customers do not wait for “the platform” to be complete because they are in production from week one.

(Volume discounts on usage-based agents and custom enterprise terms are negotiated per customer agreement.)

14. What BRAIN Is NOT

Honest accounting of limits.

14.1 BRAIN is not a black box (intentionally)

BRAIN does not use deep neural networks for its core conviction-aggregation logic. Why: opacity is the failure mode for surveillance products. A regulator examining a flagged case wants to see why; an opaque ML model can’t answer that without a long lineage of post-hoc explanation tools.

BRAIN does use ML in specific bounded contexts (within agents — e.g., ATLAS’s RUSBoost classifier, SCOUT’s bot-detection clustering). Those ML components are documented in the per-agent research foundations and have their own model cards. BRAIN’s *aggregation logic* is deliberately rule-based and traceable.

14.2 BRAIN is not a trade signal generator

BRAIN is detection. Customers do not derive trades from BRAIN outputs. The intentional design point: BRAIN’s outputs are about evidence patterns, not about market direction. A customer who tries to use BRAIN as a trade signal generator will likely lose money — manipulation detection and tradeability are different problems.

14.3 BRAIN is not a replacement for human analysts

BRAIN’s outputs are analyst inputs. Every high-conviction case requires human analyst review before regulatory reporting or other action. BRAIN’s calibration target is “surface cases worth analyst attention with reasonable precision,” not “make decisions analysts would make.”

14.4 BRAIN is not a guarantee against missed cases

False negatives exist. Sophisticated manipulators design specifically to evade detection, including detection by Vexorin-style multi-surface tools. BRAIN’s recall is calibrated against the 50-case enforcement library (documented historical cases) and synthetic non-manipulation controls; future manipulation patterns may evade BRAIN’s current taxonomy. CALIBRATE’s job is to evolve the system to catch novel patterns; this is an ongoing process.

14.5 BRAIN is not a regulator

BRAIN does not have authority to investigate, subpoena, or prosecute. It is a tool for customers (analysts, compliance teams, regulators) to make informed decisions. Outputs are evidence; investigation and action are human responsibilities.

14.6 BRAIN’s day-one performance is research-derived, not customer-tailored

The Prior Weight Matrix v1.0 weights are derived from public regulatory record (50 enforcement cases). Customer-specific calibration takes 60-90 days of CALIBRATE running. Customers expecting day-one customer-specific tailored performance will be disappointed; customers who understand the priors-to-posteriors model will experience meaningful improvement over the first 3 months.

14.7 BRAIN’s effect sizes are subject to compression

All academically-published effect sizes are subject to post-publication compression as the techniques become known. BRAIN’s per-agent weights are calibrated downward from published academic values to reflect this compression. Effect sizes in 2026 are 30-60% of original published magnitudes for most agent signals. CALIBRATE’s job is to track this compression in real-time per customer.

15. Roadmap

15.1 v0.1 (current) — Pre-paper-trade

- 10 agents + BRAIN + CALIBRATE
- Prior Weight Matrix v1.0 (research-derived priors)
- 7 manipulation families
- Self-hosted and hosted deployment options
- Native data feeds (market, news, social, filings, regulatory)
- Three customer integration models (hosted, self-hosted, air-gapped)
- Standard connectors (REST, streaming, file-drop, FIX, CDC)

15.2 v0.2 (target Q3 2026) — Post-paper-trade

- Customer-specific calibrated weights from CALIBRATE walk-forward (60-day deployments)
- 8-9 manipulation family taxonomy (extended based on observed novel patterns)
- Improved cross-agent correlation handling
- Performance optimizations (target sub-300ms end-to-end latency)

15.3 v1.0 (target 2027) — Post-live-trading

- Crypto-systemic-risk agent (using v0.1 reserved capacity)
- Behavioral-finance agent (sentiment/disposition effects)
- Regional regulator extensions (FCA, ESMA, JFSA, MAS focus)
- Federated CALIBRATE — customers can opt-in to shared learning across customers (anonymized)
- Multi-customer pattern detection (consortium-style detection of cross-firm coordination)

15.4 v2.0 (target 2028+) — Maturity

- Real-time BRAIN-of-BRAINS (cross-customer aggregation for systemic-risk applications)
- Generative-AI augmentation (for analyst report writing, case narrative generation)
- Regulator-direct deployment (white-label for regulators)
- Industry-vertical specialization (banking, energy, crypto, FX)

The roadmap is not promises; it is direction. Specifics shift based on customer demand, calibration evidence, and technical feasibility.

16. Glossary

- **Agent** — Specialized signal-detection module focused on one surface or domain (Scout, Echo, Atlas, Cipher, Nexus, Herald, Argus, Magnus, Orion, Axiom).
- **AAER** — SEC Accounting and Auditing Enforcement Release.
- **BRAIN** — The aggregation, pattern-recognition, and conviction-scoring layer.
- **CALIBRATE** — The walk-forward validation and weight-refinement layer.
- **Cascade** — Coordinated firing across multiple surfaces in canonical temporal order.
- **Conviction score** — BRAIN's 0-100 calibrated score for whether observed activity matches a manipulation family.
- **Family** — One of the 7 canonical manipulation taxonomies (pump-and-dump, short-and-distort, insider cluster, spoofing, disclosure fraud, cross-market, geopolitical/event).
- **GPR Index** — Caldara-Iacoviello Geopolitical Risk Index.
- **Manifold** — The geometric structure of signals across the six evidence surfaces.
- **Modifier** — Multiplicative adjustment to signal weights based on regime/cycle/event-window context.
- **Prior Weight Matrix v1.0** — The research-derived weight set BRAIN deploys with on day 1.
- **Surface** — One of the six nested evidence layers (Context, Intent, Positioning, Flow, Structure, Disclosure).
- **Walk-forward validation** — CALIBRATE's methodology: replay historical data and predict outcomes against subsequent labels.

17. Appendix A — Sample BRAIN Trace

Illustrative trace for a hypothetical case to show the audit-trail granularity. (Names anonymized; data illustrative.)

Case ID: 4521
Asset: ACME Corp (Ticker: ACME)
Detection Date: 2026-04-15
Conviction Score: 76 (HIGH)
Primary Family Hypothesis: Pump and Dump (match score 0.81)
Alternative Family: Coordinated Geopolitical/Event (match score 0.42)

DRIVING AGENTS (top 3):

1. SCOUT (Context surface, contribution 28%)
 - Signal: narrative_coupling fired with z-score 3.2 vs threshold 1.5
 - Sources: vosoughi_roy_aral_2018, bovet_makse_2019
 - Detail: 8,430 inauthentic accounts coordinated across Twitter/X and Telegram
primary languages: English (74%), Spanish (22%), Russian (4%)
narrative cluster: ACME bull thesis aligned with Q1 earnings expectation
2. ECHO (Positioning surface, contribution 23%)
 - Signal: takeover_options_anomaly fired with z-score 2.7 vs threshold 1.5
 - Sources: cao_chen_griffin_2005, augustin_brenner_subrahmanyam_2019
 - Detail: ATM call volume in ACME 4.2x normal, accumulated over 5 trading days
call/put ratio: 8.3 vs 90-day average 1.4
3. CIPHER (Intent surface, contribution 19%)
 - Signal: opportunistic_classification fired with classification confidence 0.78
 - Sources: cohen_malloy_pomorski_2012
 - Detail: VP-level insider sold 8,400 shares (vs his historical 10b5-1 baseline)
trade timing: 3 days before earnings; non-routine pattern

CASCADE STRUCTURE:

- Day -14: SCOUT first firing (Context)
- Day -8: CIPHER first firing (Intent)
- Day -5: ECHO first firing (Positioning)
- Day 0 (current): NEXUS pattern beginning (Flow surface, conviction 22)
- Cascade order: Context → Intent → Positioning → Flow ✓ matches Pump-and-Dump canonical

ACTIVE MODIFIERS:

- AXIOM: SYSTEM_QUIET regime detected → low-frequency signal modifier × 1.10
- ORION: No active cycle modifier
- ARGUS: No active geopolitical event window
- MAGNUS: FOMC announcement 14 days away (no modifier active yet)

CROSS-AGENT CORRELATION FACTOR: 1.05 (SCOUT/CIPHER/ECHO largely uncorrelated;
slight boost for cross-cluster firing)

PATTERN MATCH FACTOR: 1.45 (strong match against Pump-and-Dump canonical)

CASCADE FACTOR: 1.20 (clean cascade order; appropriate timing)

RAW BASE AGGREGATION: 0.42

- × Cross-agent factor 1.05 = 0.44
- × Pattern match factor 1.45 = 0.64
- × Cascade factor 1.20 = 0.77

Rounded to conviction 76

RECOMMENDATION: Surface to senior analyst
SUGGESTED EVIDENCE PACKAGE: Auto-generated with full source citations
ALERT CHANNELS: Email senior analyst; Slack #compliance-alerts

This is the level of detail customers see for any case. Drilling further: each cited source can be opened and read; each signal threshold can be inspected; each modifier rationale can be examined.

18. Appendix B — Connector Reference Matrix

Connector	Use case	Latency	Throughput	Setup complexity
REST API	Standard integration	<500ms	1000 RPS	Low
Kafka	High-volume streaming	<50ms	100K+ EPS	Medium
Kinesis	AWS-native streaming	<100ms	100K+ EPS	Medium
Azure Event Hubs	Azure-native streaming	<100ms	100K+ EPS	Medium
Pub/Sub	GCP-native streaming	<100ms	100K+ EPS	Medium
FIX 4.4/5.0	Institutional order flow	<100ms	50K+ MPS	High
FAST	High-frequency order flow	<50ms	500K+ MPS	High
S3 file-drop	Batch ingestion	Minutes	TB/day	Low
SFTP	Legacy batch	Minutes	GB/day	Low
PostgreSQL CDC	Real-time DB integration	<500ms	Variable	Medium
MySQL CDC (Debezium)	Real-time DB integration	<500ms	Variable	Medium
Snowflake/BigQuery streams	Cloud-DW integration	Minutes	TB/day	Medium
O365 Graph API	Email surveillance	Minutes	Variable	Medium
Bloomberg Vault	Compliance comms	Minutes	Variable	Medium
Symphony	Chat surveillance	Real-time	Variable	Medium
NICE Actimize	Voice + multi-channel	Variable	Variable	Medium

19. The Simulation Agent

19.1 Purpose

The Simulation Agent is the validation infrastructure for the Validated Data Marketplace (Section 20). It generates randomized synthetic manipulation scenarios faithful to the seven canonical family geometries, runs candidate third-party data through the existing agent mesh under those scenarios, and produces statistically significant validation output measuring whether the candidate data improves the mesh’s recognition of those scenarios.

The Simulation Agent solves two problems simultaneously:

- **For data providers** — the validation problem. The institutional alt-data market is fragmented because providers cannot prove signal isolation in a way buyer research teams will sign off on. The Simulation Agent produces a quantified, methodology-traceable validation artifact that providers carry into customer conversations.
- **For Vexorin’s customers** — the evaluation problem. Mid-AUM funds evaluating data sets typically burn \$200K–\$500K and 6–9 months per dataset evaluated. The Simulation Agent compresses that to a single integration test, with the validation artifact attached to the data provider’s listing in the marketplace.

19.2 Architectural components

Four components compose the Simulation Agent:

1. **Scenario generator.** Produces synthetic manipulation events parameterized from the 50-case enforcement library and the seven family centroids. Each scenario specifies which agents should fire, in what cascade order, with what signal strengths, against what synthetic market backdrop. The generator respects family geometry — a synthetic Family A event resembles GME geometrically; a synthetic Family E event resembles Iran 2026 geometrically.
2. **Agent-mesh runtime.** Scout and Carlton in production today; ECHO, ATLAS, CIPHER as the first additional agents brought online during the build phase. The runtime accepts both real and synthetic data inputs and produces signal contracts conforming to BRAIN’s aggregation protocol.
3. **Validation comparison engine.** Statistical inference layer that runs scenarios with and without the candidate provider’s data integrated, computes conviction lift across hundreds-to-thousands of scenarios per validation run, and produces statistical-significance output (effect size, p-value, confidence interval).
4. **Validation report artifact.** The customer-facing output. Structured document the data provider receives at

completion of validation, certifying mesh integration with attached methodology, confidence intervals, and replication details. The artifact is what the data provider takes into their customer conversations.

19.3 Engineering scope

The Simulation Agent is **not** a \$15K, 2-4 week build like Scout or Carlton. It depends on multiple foundational components: BRAIN aggregation as running code (not just architectural specification), additional agents brought online for meaningful mesh coverage, and the scenario generator design which is novel work without off-the-shelf precedent.

Realistic build estimate with parallel-staffed engineering: **4-5 months** to a working Simulation Agent capable of validating the first data provider. The CTO underwrites this estimate; the round funds the build.

19.4 The Simulation Agent as moat

Once operational, the Simulation Agent becomes a credentialing layer. Every validated data provider's certification is a Vexorin-branded artifact — providers want it because it is how they sell to funds; funds trust it because it is tied to a falsifiable methodology. Credentialing layers compound: each validated provider increases the marketplace's perceived rigor, which increases the next provider's willingness to validate, which increases customers' willingness to acquire from the marketplace.

20. Extension Layers — Federated Registry and Validated Data Marketplace

20.1 Two distinct extension layers

Two architecturally distinct extension layers compound on the core agent set. Both are customer-pull rather than vendor-push: nothing arrives in a customer environment without the customer browsing the registry from inside their own deployment and choosing to integrate.

- **Federated Registry** (Section 20.2) — for third-party signal modules (agent code).
- **Validated Data Marketplace** (Section 20.3) — for third-party data sets, validated through the Simulation Agent (Section 19).

20.2 Federated Registry

Independent agent developers publish MCP-compatible signal modules to a centrally-verified registry. Vexorin handles registry governance: code review, signing, version pinning, security review, MCP-protocol compliance verification. Customers browse the registry from inside their own on-prem deployment and pull verified modules into their environment when they want them. Execution stays customer-side.

Revenue economics: Vexorin takes a share of usage fees on platform consumption flowing through customer-deployed third-party modules. Specific share negotiated per developer agreement; representative range comparable to other developer marketplaces.

The Federated Registry is the architectural answer to the on-prem-vs-extensibility tension: how does an independent developer ship code into a closed customer environment? Answer: not by shipping. The developer publishes verified code to the registry; the customer pulls.

20.3 Validated Data Marketplace

Third-party data providers submit data sets to Vexorin. The Simulation Agent (Section 19) runs the candidate data through the existing mesh under randomized synthetic manipulation scenarios. Validated data sets earn placement in the marketplace with a Vexorin-branded validation artifact attached. Customers acquire data sets with proof of mesh integration — eliminating most of the conventional alt-data evaluation burn-time.

Revenue economics: - \$10,000 onboarding fee per data provider, paid at simulation analysis and deployment - 15-20% revenue share on data fees flowing through the validated marketplace to Vexorin's customer base

Pain Point 3 of the Vexorin thesis (data providers cannot prove signal isolation) is solved by this architecture: the validation artifact is the proof. The Validated Data Marketplace is what makes the third pain point of the original thesis a Vexorin revenue line rather than a customer pain point.

20.4 Extension layer governance principles

Both layers share four governance principles:

- **Customer-pull, not vendor-push.** Nothing arrives in a customer environment without the customer choosing to integrate.

- **Vexorin-verified, customer-executed.** Vexorin handles verification (code review, simulation validation, security review). Execution stays customer-side.
- **Audit-trail integrity.** Every third-party module or data set integrated into a customer deployment carries provenance metadata into BRAIN’s three-layer audit trail (Section 5.5).
- **Revenue transparency.** Both extension layers compute revenue share from auditable usage telemetry. Customers can opt into telemetry reporting (default) or reconcile via license terms (air-gapped customers).

Document Changelog

Version	Date	Changes
0.1	2026-05-04	Initial litepaper. Architectural overview, six-surface manifold explanation, conviction-aggregation mathematics, seven manipulation families, modifier system, CALIBRATE methodology, customer data integration patterns and connector reference, output surfaces, security/audit/governance, self-hosted deployment, honest limits, roadmap. Pre-paper-trade. All citations verified per v2.0 audit. Companion to per-agent research foundations (10 docs) and Prior Weight Matrix v1.0.
0.2	2026-05-07	Pricing model rewrite (Section 13.5) — replaced Starter/Growth/Enterprise tiers with three-stream model: \$50K platform fee, usage-based agent consumption (Scout \$5K/M posts, Carlton \$20/ticker, future agents per usage), and Validated Data Marketplace economics (15-20% revenue share, \$10K provider onboarding). Three-scenario marketplace projection (1%/2.5%/5% capture). Section 11.5 reframed: audit-grade evidence package for customer’s downstream regulator interactions, not a claim of Vexorin regulator engagement. Section 12.5 reframed: positioning toward customer’s regulatory scrutiny, not Vexorin’s regulator pursuit. New Section 19 (Simulation Agent) — validation infrastructure for the marketplace, 4-5 month CTO-underwritten build. New Section 20 (Extension Layers) — Federated Registry for third-party signal modules and Validated Data Marketplace for third-party data sets, customer-pull architecture, Vexorin-verified governance.

End of BRAIN litepaper v0.1.